



Hetvi Udaybhai Desai

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ABOUT ME

I am a motivated IT student with a strong academic foundation and hands-on experience through academic projects and research activities. Demonstrates strong problem-solving abilities, technical aptitude, and a proactive approach to applying theoretical knowledge to real-world scenarios.

WORK EXPERIENCE

AI/ML ENGINEER (INTERN) – GURUGRACE CONSULTANCY SERVICE PVT. LTD (REMOTE) – 2025/12/15 – 2026/06/30 – AHMEDABAD, INDIA

Business or Sector: Information and communication | **Department:** Natural language processing |

Address: D/803, Sagar Samrat Flat, Times of India Roads, Vejalpur, Ahmedabad, Gujarat, 380051, Ahmedabad, India | **Email:** info@gcsteg.com |

Website: <https://gcsteg.com/>

- Architected an emotionally aligned AI ecosystem utilizing prompt engineering, activation steering, and circuit-level interventions; achieved state-of-the-art emotion-expression accuracy of 97.4% on Mistral-3-3B and 99.4% on Qwen3-4B-Instruct.
- Developed an end-to-end multimodal emotion pipeline integrating a RoBERTa classifier (98.8% F1), target classifier (96.6% F1), and a LoRA-fine-tuned SenseVoice model for speech recognition. Combined these with CosyVoice2 and F5-TTS to enable controllable emotional speech generation across 6+ distinct categories.

RESEARCH INTERN – INDIAN INSTITUTE OF TECHNOLOGY, HYDERABAD, NLIP: NATURAL LANGUAGE AND INFORMATION PROCESSING LAB – 2026/01/26 – 2026/05/23 – HYDERABAD, INDIA

Business or Sector: Information and communication | **Department:** NLIP: Natural Language and Information Processing Lab |

Email: nlip@csse.iith.ac.in | **Website:** <https://nlip-lab.github.io/nlip/>

- This project focuses on localizing school-level math word problems and kids stories from US/UK English to Indian English while preserving identical mathematical meaning and structure.
- It builds a high-quality synthetic parallel dataset using open-source large language models with strict structural alignment constraints.
- The generated dataset is then used to train and evaluate localization models for controlled dialect adaptation and educational NLP applications.

RESEARCH INTERN – INDIAN INSTITUTE OF TECHNOLOGY (IIT) MANDI , CENTRE FOR ARTIFICIAL INTELLIGENCE AND ROBOTICS (CAIR) – 2025/05/26 – 2025/08/20 – MANDI, INDIA

Business or Sector: Information and communication | **Department:** Artificial Intelligence and Robotics | **Email:** cairoffice@iitmandi.ac.in |

Website: <https://research.iitmandi.ac.in/cair/>

- Developed humanoid locomotion behaviors using Imitation Learning techniques such as GAIL and TRPO for human-like gait synthesis.
- Processed and analyzed motion capture data to learn state-action policies directly from expert demonstrations, enabling realistic movement generation.
- Implemented and trained Reinforcement Learning models (SAC, DDPG, TD3) for bipedal robot walking, achieving stable and robust gait control in simulation.

EDUCATION AND TRAINING

2022/09/06 – 2026/04/28 Anand, India

BACHELOR OF TECHNOLOGY (BTECH) INFORMATION TECHNOLOGY Charotar University of Science and Technology (CHARUSAT)

During my bachelor's degree, I completed core coursework in Programming Languages, Data Structures, Database Management Systems, Operating Systems, Computer Organization and Architecture, Design of Algorithms, Computer Networks, Software

Engineering, Theory of Computation, Information Security, Cloud Computing, Compiler Construction, Machine Learning, Deep Learning, and Artificial Intelligence.

Address CHARUSAT Campus, Off. Nadiad-Petlad Highway, Changa, 388421, Anand, India | **Website** <https://www.charusat.ac.in/> |

Field of study Information Technology | **Final grade** 9.36

2020/05/10 – 2022/05/05 Kapadwanj, India
HIGHER SECONDARY SCHOOL Jeevanshilp International School

Address Jeevanshilp Campus, Garod Road, Near Gayatri Society, 387620, Kapadwanj, India | **Website** <http://jeevanshilp.org/> |

Field of study Education science , Physics, Chemistry, Maths | **Final grade** 70.46%

2019/06/09 – 2020/05/11 Kapadwanj, India
SECONDARY SCHOOL Jeevanshilp International school

Address Jeevanshilp Campus, Garod Road, Near Gayatri Society, 387620, Kapadwanj, India | **Website** <http://jeevanshilp.org/> |

Field of study Education | **Final grade** 79%

● **LANGUAGE SKILLS**

Mother tongue(s): **GUJARATI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	B2	C1	C1	B2
HINDI	C1	C1	C1	C1	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **PROJECTS**

2025/05/29 – 2025/08/21
Learning to Walk: Bipedal Robot Locomotion Using Reinforcement Learning

- Bipedal locomotion is a challenging robotics problem due to the need for dynamic balance, stability, and energy efficiency in complex control environments.
- This project uses Deep Reinforcement Learning with Actor–Critic algorithms (DDPG, TD3, SAC) and a PD controller to learn stable, robust walking in continuous action spaces.
- Trained in simulation with sensory feedback, the model achieves smooth, energy-efficient, and terrain-adaptive humanoid locomotion.

2025/02/03 – 2025/04/14
Cold Email Generator – an End-to-End Generative AI Tool

- Built a full-stack Generative AI solution to automate personalized cold email outreach for software and AI service companies.
- Leveraged LLaMA 3.1 with Retrieval-Augmented Generation (RAG) using ChromaDB and LangChain to generate context-aware, tailored emails.
- Enabled a user-friendly frontend using Flask, allowing users to input industry, service offerings, and tone preferences for generating relevant emails.

2024/09/16 – 2024/11/29
Generative AI-based SQL chatbot

- Engineered an AI-powered SQL Query Generator Chatbot, enabling users to create complex SQL queries from natural language input, improving query efficiency by 60%.
- Integrated machine learning algorithms to interpret user intent and convert it into precise SQL queries, enhancing database interaction for 50+ queries daily. Refined the AI model with continuous training on 10,000+ query variations, boosting accuracy and user experience.

● **SKILLS**

Artificial Intelligence | Data Science | Natural Language Processing | Speech Processing | Computer Vision | Machine Learning | Deep Learning | Jupyter Notebook | Large Language Model | GenAI | Agentic AI | Convolutional Neural

Networks | Image Processing | Data Preprocessing | Data Visualization | Reinforcement Learning | Object-Oriented Programming | Data Analysis

Technical

Docker | Tableau | Numpy | MySQL | React.js | Git | Express.js | Node.js | Orange | Spacy | Keras | Power BI | MongoDB | SQL

Non-Technical

Time management | Analytical Thinking | teamwork | critical thinking | communication | Leadership

Programming Languages

Python | C++ | java

AI/ML Frameworks

Pandas | OpenCV | Mujoco | Scikit-Learn | PyBullet | TensorFlow | PyTorch | Seaborn | Gymnasium | Matplotlib | Imitation Learning | NLTK | Stable-Baselines3 | Streamlit

PUBLICATIONS

2025
[Symptom-based early detection and classification of plant diseases using AI-driven CNN+ KNN Fusion Software \(ACKFS\)](#)

Jayswal Hardik , Rishi Sanjaykumar Patel b , Hetvi Desai , Hasti Vakani , Mithil Mistry , Nilesh Dubey

Authors: Jayswal Hardik , Rishi Sanjaykumar Patel b , Hetvi Desai , Hasti Vakani , Mithil Mistry , Nilesh Dubey | **Journal Name:** Software Impacts | **Publisher:** Elsevier B.V

2025
[Plant Disease Classification with Spectral Signature Taxonomy and Analysis Software \(SSTAS\)](#)

Jayswal Hardik A., Hetvi Desai, Hasti Vakani, Mithil Mistry, Nilesh Dubey et al. Software Impacts, Accepted for publication in Software Impacts (Web of Science - ESCI, Impact Factor: 1.3, Q3).

Authors: Jayswal Hardik A., Hetvi Desai, Hasti Vakani, Mithil Mistry, Nilesh Dubey et al | **Journal Name:** Software Impacts | **Publisher:** Elsevier B.V

2024
[Crop Recommendation Systems: Insights, Trends, and Methodological Approaches](#)

Hetvi Desai, Hardikkumar Jayswal, Nilesh Dubey, Dipika Damodar, Ashwin Makwana, Jitendra Chaudhari & Amit Nayak

Authors: Hetvi Desai, Hardikkumar Jayswal, Nilesh Dubey, Dipika Damodar, Ashwin Makwana, Jitendra Chaudhari & Amit Nayak | **Journal Name:** International Conference on Information and Communication Technology for Competitive Strategies | **Volume, Issue and Pages:** Intelligent Strategies for ICT,LNNS,volume 1319, pp 351–359 | **Publisher:** Springer Nature Singapore Pte Ltd

VOLUNTEERING

2023/01/07 – 2025/01/24 CHARUSAT UNIVERSITY, CHANGA
Executive Central Council - University Level Student Council

Actively organize and manage university-level sports, cultural, and technical events, ensuring seamless execution for 10,000+ students. Responsible for event planning, coordination, and overall management to enhance student engagement.

2024/09/20 – 2024/09/24 CHARUSAT UNIVERSITY, CHANGA
TED X CHARUSAT

Organizing Team Member at TEDx CHARUSAT, responsible for speaker hospitality, coordination, and on-site support to ensure a smooth and professional event experience.